# Block practice · More maths skills for biology data

Work through each question type in order. Each section has a worked-example retry and all further practice. Attempt only the part outlined in red.

Convert between pie-chart sectors, percentages and counts

Scale a rate per 100 000 people to a whole population

Estimate an unmeasured value by interpolating or extrapolating data

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# Practice workbook · Convert between pie-chart sectors, percentages and counts

A pie chart or circular cycle diagram is read, completed or drawn, with sectors as angles, percentages or time fractions.

4 question parts: 1 worked-example retry and 3 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-MATH-PIE-CHART-SECTORS

# 1. Question 8 · part b.i · 3 marks

June 2025 Paper 2R

Worked-example retry · Attempt part b.i only.

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**8** (a) Read the passage below.

Use the information in the passage and your own knowledge to answer the questions that follow.

The human body has physical and chemical barriers that reduce the risk of pathogens entering the body. These barriers are known as the first line of defence. If the barriers are penetrated, then the second line of defence comes into action.

The cells involved in the second line of defence form part of the non-specific line of defence. They do not target specific pathogens but they play an important role in stimulating cells involved in the next stages of defence against disease.

The third line of defence involves specific immune responses by cells. These cells rely on the presence and detection of marker chemicals found on the surface of microscopic invaders. The cells of the specific immune system have receptor proteins that interact with the marker chemicals. The marker chemicals cause the cells to become activated and produce antibodies. The production of antibodies in this way is known as natural, active immunity and results in long-term immunity. However, there are other ways that antibodies can provide protection against disease. Natural passive immunity and artificial passive immunity are two similar processes that minimise the risk of too much harm being caused by pathogens in the body.

(i) Name a physical barrier that prevents the entry of pathogens into the body. (lines 1 to 3)

**(1)**

(ii) Suggest how chemical barriers can reduce the harm caused to the body by pathogens. (lines 1 to 3)

**(1)**

(iii) Phagocytes are one type of cell involved in the second line of defence.

Describe the process of phagocytosis. (lines 4 to 6)

**(2)**

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# 1. Question 8 · part b.i · 3 marks

June 2025 Paper 2R

Worked-example retry · Attempt part b.i only.

(iv) State the scientific name for the marker chemicals found on the surface of microscopic invaders. (lines 7 to 11)

**(1)**

(v) Suggest what is meant by a specific immune response. (line 7)

**(2)**

(vi) Describe the differences between natural passive immunity and artificial passive immunity. (lines 14 to 16)

You should give examples in your answer.

**(2)**

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# 1. Question 8 · part b.i · 3 marks

June 2025 Paper 2R

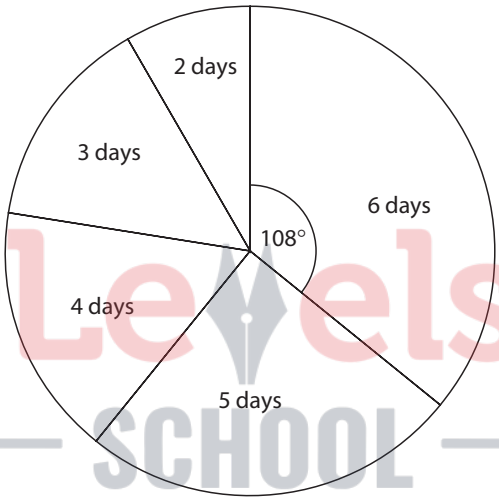
Worked-example retry · Attempt part b.i only.

(b) Scientists test how quickly 30 different types of antibiotic reduce the population size of a resistant strain of bacteria.

The scientists record the number of antibiotics that reduce the population of the resistant strain of bacteria by half.

They do this after 2 days, 3 days, 4 days, 5 days or 6 days.

Each section of the pie chart represents the number of antibiotics.




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(i) Calculate the number of types of antibiotic that reduce the population size of the resistant strain of bacteria by half in 6 days.

**(3)**

number of types of antibiotic = .....

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# 1. Question 8 · part b.i · 3 marks

June 2025 Paper 2R

Worked-example retry · Attempt part b.i only.

(ii) State a reason why it is important to wear gloves and a face mask when investigating bacteria.

**(1)**

(iii) Describe how the use of antibiotics can cause antibiotic-resistant strains of bacteria.

**(2)**

**(Total for Question 8 = 15 marks)**

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Working and answer

## 2. Question 4 · part a.i · 2 marks

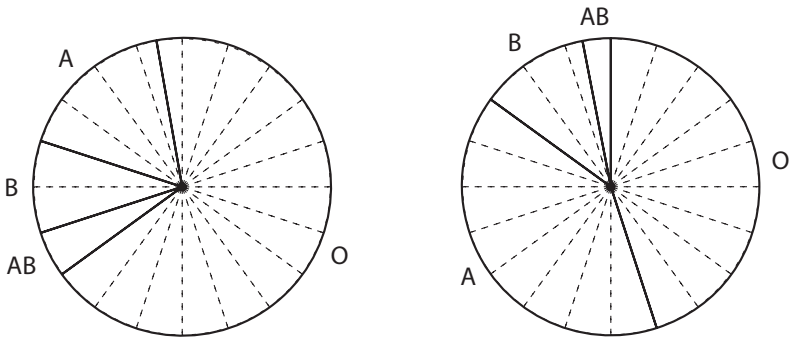
June 2022 Paper 1

Further practice · Attempt part a.i only.

**4** (a) Humans belong to one of four blood groups.

These blood groups are A, B, AB and O.

The pie charts show estimates of the percentages of people from Africa and the percentages of people from Europe in each blood group.




(i) Complete the tables to give the percentage of each blood group.

Some information has been given.

**(2)**

**Africa**

| Blood group | Percentage of blood group |
|---|---|
| O | 68 |
| A | |
| B | |
| AB | 5 |

**Europe**

| Blood group | Percentage of blood group |
|---|---|
| O | 45 |
| A | |
| B | 12 |
| AB | |

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## 2. Question 4 · part a.i · 2 marks

June 2022 Paper 1

Further practice · Attempt part a.i only.

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(ii) Some people need a blood transfusion.

Explain why people with blood group A cannot receive a blood transfusion from people with blood group B.

**(3)**

(iii) A male with the genotype $I^AI^O$ and a female with the genotype $I^BI^O$ have a child with the genotype $I^OI^O$.

Draw a genetic diagram to show how the child inherits the genotype $I^OI^O$.

**(3)**

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(iv) Determine the probability that a child from these parents will inherit the blood group AB.

**(1)**

probability = .....

## 2. Question 4 · part a.i · 2 marks · Working and answer

Working and answer

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## 3. Question 6 · part a.i · 2 marks

November 2020 Paper 1R

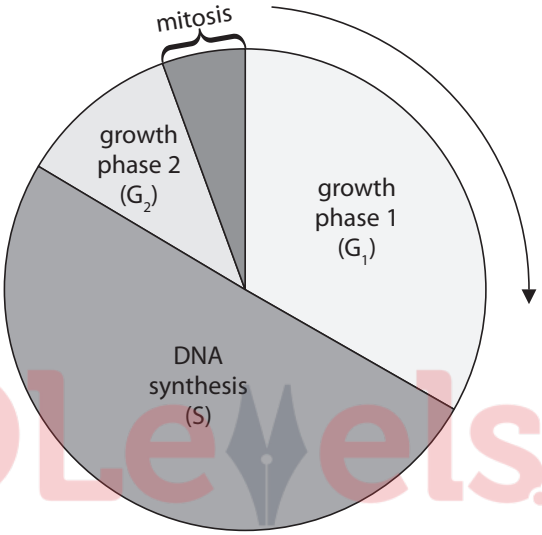
Further practice · Attempt part a.i only.

**6** (a) Cells in the body go through several cell cycles in their lifetime.

Each cell cycle is divided into different stages.

Different events take place in the cell during each stage.

The diagram shows one cell cycle of a body cell.




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(i) One cell cycle takes 24 hours to complete.

The total time spent in $G_1$ and $G_2$ is ten hours.

Calculate the time spent in mitosis.

**(2)**

time = ..... hours

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## 3. Question 6 · part a.i · 2 marks

November 2020 Paper 1R

Further practice · Attempt part a.i only.

(ii) Each of the phases of the cell cycle prepare the cell for mitosis.

Suggest the events that take place in the growth phases of the cell cycle.

**(2)**

(iii) The cell cycle completes by producing two new body cells.

Describe the genetic features of the two cells produced by mitosis.

**(2)**

(iv) The table lists some of the stages of mitosis.

Complete the table by giving details of the event that takes place in each stage.

**(3)**

| Stage | Events |
|---|---|
| metaphase | |
| anaphase | |
| telophase | |

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## 3. Question 6 · part a.i · 2 marks · Working and answer

Working and answer

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# 4. Question 4 · part a · 4 marks

June 2023 Paper 1R

Further practice · Attempt part a only.

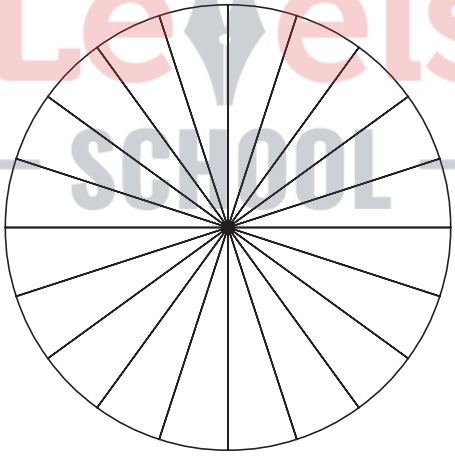
**4** The table shows the percentage composition of the diets of two 10 year-old girls.

One girl lives in Area A and the other girl lives in Area B.

| Component | Percentage (%) composition of diet | |
|---|---|---|
| | Area A | Area B |
| cereals | 75.0 | 15.0 |
| fruit and vegetables | 15.0 | 37.5 |
| eggs, fish and meat | 2.5 | 27.5 |
| milk, cheese and butter | 7.5 | 20.0 |

(a) Complete the pie chart to show the diet of the girl in Area B.

**(4)**



(b) Explain what is meant by a balanced diet.

**(2)**

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## 4. Question 4 · part a · 4 marks · Working and answer

Working and answer

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# Practice workbook · Scale a rate per 100 000 people to a whole population

Disease cases or deaths are given, often read from a graph, per 100 000 people; the number in a stated population is required.

5 question parts: 1 worked-example retry and 4 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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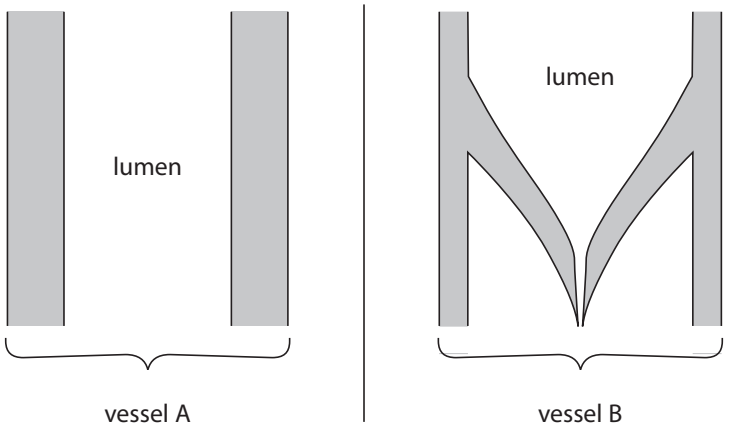
Type ID: HBI-MATH-RATE-PER-100000-SCALING

# 1. Question 3 · part b.ii · 3 marks

June 2022 Paper 2R

Worked-example retry · Attempt part b.ii only.

**3** (a) The diagram shows two types of blood vessel, A and B.




(i) Draw an arrow on the diagram to show the direction of blood flow in vessel B.

**(1)**

(ii) Identify each type of blood vessel, giving explanations for why there are differences in their structures.

**(4)**

Vessel A

Vessel B

(iii) State the name of one other type of blood vessel.

**(1)**

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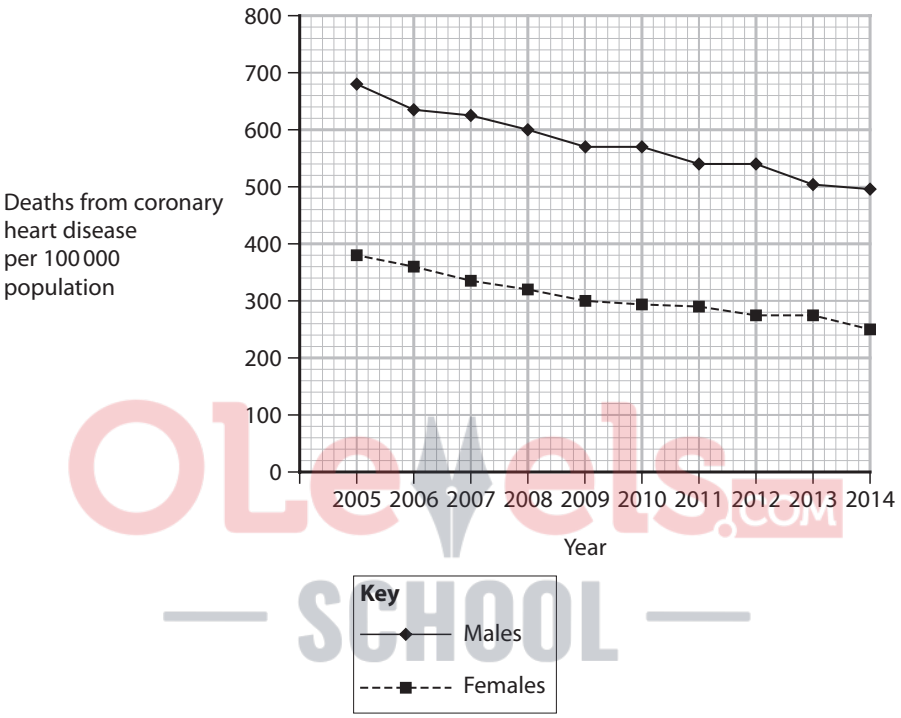
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# 1. Question 3 · part b.ii · 3 marks

June 2022 Paper 2R

Worked-example retry · Attempt part b.ii only.

(b) Coronary heart disease kills many people each year. The graph shows the number of deaths per 100 000 of the whole population in a part of the world.




(i) Describe the trends shown by the graph.

**(3)**

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# 1. Question 3 · part b.ii · 3 marks

June 2022 Paper 2R

Worked-example retry · Attempt part b.ii only.

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(ii) The part of the world for which the data was collected had a population of 60 million people in 2006.

Calculate the number of men who died from coronary heart disease in 2006.

**(3)**

number of men who died = .....

**(Total for Question 3 = 12 marks)**

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Working and answer

## 2. Question 6 · part a.iii · 2 marks

January 2020 Paper 2

Further practice · Attempt part a.iii only.

**6** *Staphylococcus aureus* (*S. aureus*) and MRSA are microorganisms that cause a wide range of infections in humans.

The table shows the number of deaths from *S. aureus* and from MRSA in England in one year.

The data is for males and females of different age groups.

| Age group in years | Number of deaths in one year per million people | | | |
|---|---|---|---|---|
| | Males | | Females | |
| | ***S. aureus*** | **MRSA** | ***S. aureus*** | **MRSA** |
| under 45 | 1.8 | 0.5 | 1.4 | 0.4 |
| 45–54 | 6.0 | 2.5 | 4.3 | 2.4 |
| 55–64 | 15.7 | 8.2 | 8.1 | 4.2 |
| 65–74 | 41.2 | 26.1 | 21.9 | 12.5 |
| 75–84 | 128.6 | 86.8 | 81.1 | 51.6 |
| 85 and over | 416.6 | 299.9 | 200.3 | 146.4 |

Source: *Office for National Statistics*

(a) (i) The data shows that there are more deaths caused by *S. aureus* than by MRSA.

Give two other conclusions that can be made from the data.

**(2)**

1 .....

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2 .....

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## 2. Question 6 · part a.iii · 2 marks

January 2020 Paper 2

Further practice · Attempt part a.iii only.

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(ii) Calculate the ratio of males aged 85 and over who die from *S. aureus* infection to males under the age of 45 who die from *S. aureus* infection.

**(2)**

ratio = .....

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(iii) The total population in England is 53 million.

Calculate the number of people aged between 65 and 74 who died from *S. aureus*.

**(2)**

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number = .....

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(iv) The number of deaths caused by MRSA is lower than the number of deaths caused by *S. aureus*.

Explain why an infection caused by MRSA is still a major concern.

**(2)**

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## 2. Question 6 · part a.iii · 2 marks · Working and answer

Working and answer

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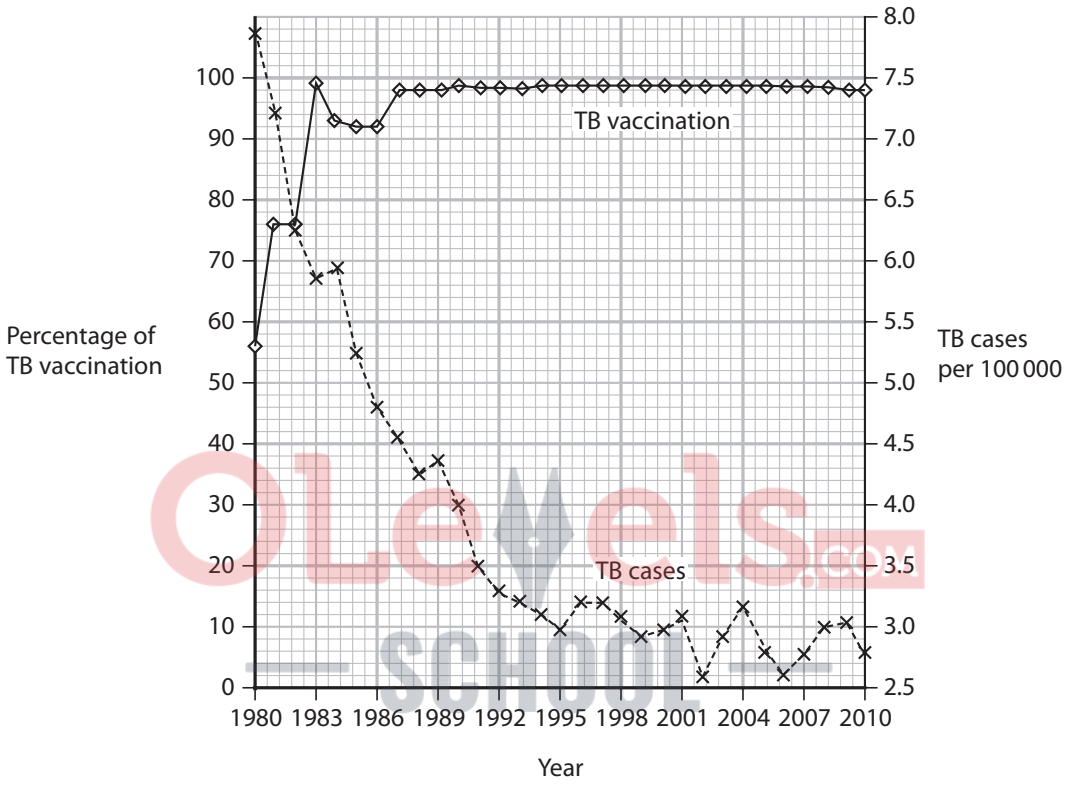
# 3. Question 4 · part b · 3 marks

June 2021 Paper 1

Further practice · Attempt part b only.

**4** The graph shows the percentage of people vaccinated against tuberculosis (TB) in a country between 1980 and 2010.

It also shows the number of cases of tuberculosis in that country.




(a) (i) Tuberculosis is caused by the same type of organism as typhoid.

What is the type of organism?

**(1)**

- ☒ **A** bacterium
- ☒ **B** fungus
- ☒ **C** protozoan
- ☒ **D** virus

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## 3. Question 4 · part b · 3 marks

June 2021 Paper 1

Further practice · Attempt part b only.

(ii) Describe the relationship between the percentage vaccination and the number of cases of TB.

**(4)**

(b) The population of this country in 2010 was 3.45 million.

Calculate the number of cases of tuberculosis in this country in 2010.

**(3)**

number of cases = .....

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## 3. Question 4 · part b · 3 marks

June 2021 Paper 1

Further practice · Attempt part b only.

(c) Explain how vaccination can give a person immunity to a disease.

**(5)**

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**(Total for Question 4 = 13 marks)**

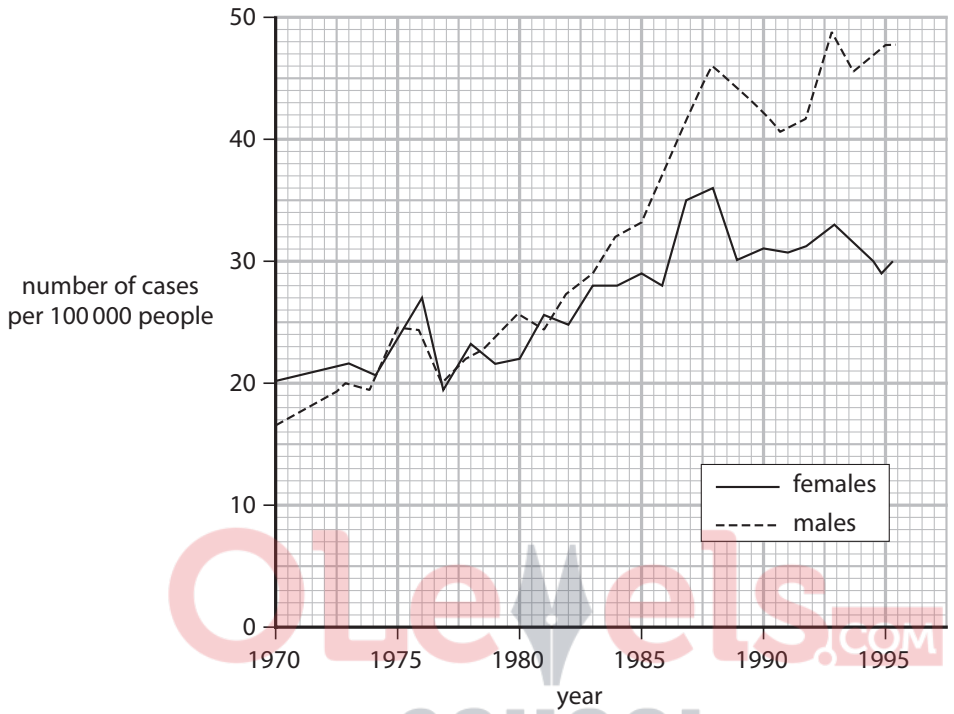
Working and answer

# 4. Question 6 · part b.ii · 3 marks

November 2021 Paper 1

Further practice · Attempt part b.ii only.

**6** A form of skin cancer called melanoma has been increasing in both males and females.




(a) (i) Describe the trends shown by the graph.

**(3)**

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# 4. Question 6 · part b.ii · 3 marks

November 2021 Paper 1

Further practice · Attempt part b.ii only.

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(ii) Suggest why cancer can occur in the skin.

**(2)**

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(b) (i) Calculate the mean increase in the number of melanoma cases per 100 000 males per year between 1970 and 1995.

**(3)**

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mean increase per 100 000 males per year = .....

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(ii) In 1970, the population of this area of the world was 7 million.

Calculate the number of females who had melanoma.

**(3)**

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number of females with melanoma = .....

**(Total for Question 6 = 11 marks)**

## 4. Question 6 · part b.ii · 3 marks · Working and answer

Working and answer

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# 5. Question 4 · part c.ii · 3 marks

November 2024 Paper 2

Further practice · Attempt part c.ii only.

**4** Read the passage below.

Use the information in the passage and your own knowledge to answer the questions that follow.

When looking at data, it is clear that people in lower income countries are more likely to die from infectious diseases (approximately 240 per 100 000) than non-infectious diseases (approximately 180 per 100 000). Infectious diseases such as diarrhoea and malaria can be easily eliminated or treated. It is also possible to vaccinate against diseases such as TB. These diseases are uncommon in higher income countries because of preventative medicine.

The only common diseases present in both higher and lower income countries are respiratory tract infections. These infections are usually influenza, pneumonia and bronchitis. Whilst influenza is a viral infection, the other infections are usually bacterial and can be treated. Many people at risk in the higher income countries are encouraged to get preventative treatment. These are often provided free of charge for the elderly who are most at risk and those who work in indoor spaces such as schools, offices and hospitals. The effect on mortality in lower income countries is more than two times as great as in higher income countries.

The greatest killers in higher income countries are heart disease and stroke. These two diseases kill approximately twice as many people per 100 000 as in lower income countries. These are often seen as lifestyle diseases. When we compare the number of deaths caused by non-infectious diseases, in higher income countries there are 445 deaths per 100 000 and in lower income countries there are 185 deaths per 100 000. There are also more cases of other diseases such as dementia and cancers in higher income countries.

(a) State one way the infectious diseases diarrhoea and malaria can be prevented (lines 3 and 4).

**(2)**

diarrhoea

malaria

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# 5. Question 4 · part c.ii · 3 marks

November 2024 Paper 2

Further practice · Attempt part c.ii only.

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(b) (i) State how pneumonia and bronchitis can be treated (lines 8 to 10).

**(1)**

(ii) Give one reason why elderly people and people who work in indoor spaces are at risk from respiratory tract infections. (lines 12 and 13).

**(2)**

elderly people

indoor spaces

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(c) (i) Give two factors that may contribute to lifestyle diseases (line 18).

**(2)**

1 .....

2 .....

(ii) In a higher income country of 5 000 000 people, calculate the number of people who will have non-infectious diseases (lines 20 and 21).

Give your answer to three significant figures.

**(3)**

number of people = .....

## 5. Question 4 · part c.ii · 3 marks

November 2024 Paper 2

Further practice · Attempt part c.ii only.

(iii) Calculate the ratio of the number of people with non-infectious diseases in a higher income country to the number of people with non-infectious diseases in a lower income country (lines 20 and 21).

Give your answer in the form n : 1

**(3)**

ratio = ..... : .....

(iv) Suggest why there are more cases of dementia and cancers in higher income countries (lines 21 and 22).

**(2)**

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**(Total for Question 4 = 15 marks)**

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## 5. Question 4 · part c.ii · 3 marks · Working and answer

Working and answer

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Practice workbook · Estimate an unmeasured value by interpolating or extrapolating data

Data give values on either side of the target, or a steady trend that can be extended, and a value not measured is to be estimated.

4 question parts: 1 worked-example retry and 3 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-MATH-ESTIMATE-INTERPOLATE-EXTRAPOLATE

# 1. Question 4 · part b · 4 marks

June 2025 Paper 2

Worked-example retry · Attempt part b only.

**4** A student investigates the effect of different concentrations of sugar solution on the mass of cylinders made from potato.

This is the student's method.

- push a cork borer into different parts of a potato
- remove five cylinders of potato
- wrap each cylinder in a piece of filter paper
- remove from the filter paper and find the mass of each cylinder
- place one cylinder into distilled water
- place each of the other cylinders into different concentrations of sugar solution
- remove the cylinders after two hours and wrap in filter paper
- remove from the filter paper and find the new mass

These are the student's results.

| | | |
|---|---|---|
| 1.00 au = −7 % | 1.50 au = −10 % | 0.00 au (distilled water) = +8 % |
| | 0.25 au = +4 % | 0.50 au = −1 % |

The student's results are given as the percentage change in mass of the cylinder and the concentration of sugar is given in arbitrary units (au).

(a) (i) Draw a table to display the results.

**(4)**

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P 7 8 7 6 9 A 0 8 2 0

# 1. Question 4 · part b · 4 marks

June 2025 Paper 2

Worked-example retry · Attempt part b only.

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(ii) Determine the median value for sugar concentration.

**(1)**

median value = .....

(iii) The original mass of the potato cylinder placed into the sugar solution, of concentration 0.25 au, was 7.0 g.

Calculate the mass of the cylinder after two hours.

Give your answer to two significant figures.

**(3)**

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mass of cylinder = ..... g

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(b) Use the student's results to estimate the concentration of sugar solution that would produce no change in the mass of the potato cylinder.

You should include details of how you obtained your answer.

**(4)**

concentration of sugar solution = ..... au

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P 7 8 7 6 9 A 0 9 2 0

# 1. Question 4 · part b · 4 marks

June 2025 Paper 2

Worked-example retry · Attempt part b only.

(c) (i) Explain why filter paper was wrapped around the potato cylinders at the beginning and end of the investigation.

**(2)**

(ii) The student kept the potato cylinders in the liquid for two hours.

State two other factors that should be kept constant during this investigation.

**(2)**

1 .....

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2 .....

.....

**(Total for Question 4 = 16 marks)**

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## 1. Question 4 · part b · 4 marks · Working and answer

Working and answer

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## 2. Question 4 · part b.i · 1 mark

June 2022 Paper 1R

Further practice · Attempt part b.i only.

**4** (a) Lipids are digested in the small intestine.

(i) Describe a test for lipids in food.

**(3)**

(ii) Name the enzyme that digests lipids.

**(1)**

(iii) Name the products of lipid digestion.

**(2)**

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(b) A student investigates how the pH of a lipid solution, containing an enzyme that digests lipids, changes over 10 minutes at two different temperatures.

The tables show the student's results.

**Temperature at 37 °C**

| Time in minutes | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| pH | 8.5 | 8.0 | 7.5 | 7.2 | 7.0 | 6.8 | 6.2 | 5.8 | 5.5 | 5.5 |

**Temperature at 80 °C**

| Time in minutes | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| pH | 9.2 | 9.0 | 8.8 | 8.8 | | 8.6 | 8.6 | 8.4 | 8.4 | 8.4 |

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## 2. Question 4 · part b.i · 1 mark

June 2022 Paper 1R

Further practice · Attempt part b.i only.

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(i) Estimate a pH value for the lipid solution at 80 °C at 5 minutes.

**(1)**

(ii) Give the independent variable and the dependent variable for this investigation.

**(2)**

independent

dependent

(iii) Explain the differences in the results of the student's investigation at the two different temperatures.

**(4)**

**(Total for Question 4 = 13 marks)**

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## 2. Question 4 · part b.i · 1 mark · Working and answer

Working and answer

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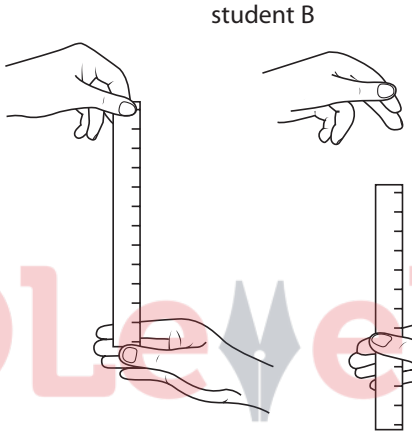
## 3. Question 5 · part e · 1 mark

November 2021 Paper 2

Further practice · Attempt part e only.

**5** Two students use this method to investigate if caffeine affects the speed of a reflex action.

- student A sits in a chair holding out a hand with their thumb and forefinger open
- student B holds a metre rule with the 0 mm mark just above the open thumb and forefinger of student A
- student B releases the ruler
- student A catches the ruler as quickly as possible by closing their thumb and forefinger




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The method is repeated 10 times.

The student catching the ruler now drinks a cup of coffee and waits for 30 minutes.

The investigation is then repeated.

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## 3. Question 5 · part e · 1 mark

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The table shows the student's results.

| Conditions \ Trial | Distance travelled by ruler in mm | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | mean |
| without coffee | 210 | 190 | 200 | 205 | 207 | 195 | 198 | 195 | 199 | 201 | 200 |
| with coffee | 165 | 160 | 163 | 167 | 165 | 170 | 175 | 170 | 180 | 175 | |

(a) Complete the table by calculating the missing mean value.

**(2)**

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(b) Coffee contains caffeine.

State the effect of caffeine on the reflex action.

**(1)**

(c) Explain why the student waited 30 minutes after drinking the coffee before continuing with the investigation.

**(3)**

## 3. Question 5 · part e · 1 mark

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Further practice · Attempt part e only.

(d) The student who catches the ruler does not eat or drink anything for several hours before the investigation.

Suggest why this improves the validity of the investigation.

**(2)**

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(e) A cup of coffee contains 100 mg of caffeine. A can of cola contains 50 mg of caffeine.

If the student catching the ruler had drunk a can of cola instead of coffee, estimate the mean distance travelled by the ruler.

**(1)**

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mean distance = ..... mm

**(Total for Question 5 = 9 marks)**

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Working and answer

# 4. Question 7 · part a.iii · 2 marks

June 2023 Paper 1

Further practice · Attempt part a.iii only.

**7** A method that shows whether a person's kidneys are working correctly is to measure the volume of filtrate produced by the kidneys in one minute. This is known as the glomerular filtration rate (GFR).

A person with healthy kidneys has a GFR of over 100 $cm^3$ per minute.

The GFR of a person whose kidneys were not working correctly was measured once a year for nine years.

The table shows the results.

| **Year** | 2011 | 2012 | 2013 | 2014 | 2015 | 2016 | 2017 | 2018 | 2019 |
|---|---|---|---|---|---|---|---|---|---|
| **GFR in $cm^3$ per minute** | 80 | 75 | 79 | 79 | 67 | 60 | 50 | 45 | 35 |

(a) (i) Plot a line graph of the data on the grid.

**(4)**



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# 4. Question 7 · part a.iii · 2 marks

June 2023 Paper 1

Further practice · Attempt part a.iii only.

(ii) Explain why it is not possible to determine from the data when the person's kidneys stopped working correctly.

**(2)**

(iii) A person needs dialysis if the GFR falls to 15 $cm^3$ per minute.

Use the data to estimate when the person will probably need to start dialysis.

**(2)**

(iv) Discuss the accuracy of this estimate.

**(3)**

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## 4. Question 7 · part a.iii · 2 marks · Working and answer

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